When Does Deep Reinforcement Learning Help Link Adaptation? A Pre-Registered Evaluation Against Calibrated Rules

Author names (to be completed)

***Abstract*—We study downlink link adaptation (CQI-offset and MCS selection) in a simulated multi-user LTE/NR-like cell with multi-process HARQ. We ask when deep reinforcement learning (RL) beats not only outer-loop link adaptation (OLLA) but also deterministic rules calibrated on the learner's own cues. The protocol uses pre-registered decision rules and wording, disjoint calibration and evaluation traces, common random numbers, seed-level inference and a public registry of 67 cases. For deadline traffic at an HARQ round-trip time of 8 TTIs, a residual advantage actor-critic (A2C) policy delivers 7.49 % more packets on time than OLLA after a 200-TTI warm-up. Against rules re-calibrated on 500 fresh traces the picture changes: re-calibration removes 58 % of the margin over the best rule. The policy still beats the strongest rule, a table that combines deadline slack, user class and HARQ-process state, on evaluation traces (+0.66 %), but not robustly across training seeds (5 of 10 seeds above zero) and not after correction over all tests in the project. A residual policy on a non-oracle base rule, found in a design ablation and confirmed in a pre-registered replication on fresh traces and seeds, beat the same combined table at trace level and across training seeds (+3.35 %, hierarchical 95 % CI [2.22, 4.40] %), although that table uses the true speed class and the policy does not. The oracle-base recipe did not reproduce at two neighbouring operating points, but the non-oracle recipe did: there it beat the OLLA family and the single-cue rules re-calibrated for each point (+1.73 % and +1.02 % over the best rule); the combined table was pre-registered at the M1 point only, and two further grid corners were not tested. The gain is also uneven: for users slower than 10 m/s the margin over a slack-indexed table is negative (−1.71 %). On full-buffer goodput the learned policy gives no gain (−0.88 % against a calibrated offset). We document three simulation pitfalls that had produced large spurious gains, and release code, traces and the registry.**

***Index Terms*—Link adaptation, outer-loop link adaptation, reinforcement learning, HARQ, evaluation methodology, reproducibility, pre-registration**

I. Introduction

Link adaptation (LA) chooses the modulation and coding scheme (MCS) for each downlink transmission from channel quality indicator (CQI) reports that are delayed and noisy. In LTE and 5G New Radio the industrial baseline is outer-loop link adaptation (OLLA) [1]: an additive offset on the reported CQI that steps down after a NACK and up after an ACK, so that the block-error rate tracks a target. Many recent studies replace or augment OLLA with bandits or deep reinforcement learning (RL) and report clear gains in throughput or reliability over it [2], [3], [4], [5], [6], [7].

Draft v3. Items marked as placeholders wait on runs or on later drafting.

A learner usually observes more than OLLA does: deadline slack, delay history, buffer or HARQ state. A gain over OLLA therefore mixes two things, the value of the extra information and the value of *learning* a policy from it. Only a comparator that receives the same information, through a rule calibrated on the learner's cues, separates the two. Of the 23 learning-based LA papers we screened (abstracts and accessible text), 0 compared against such a rule, 3 did so partly, 13 did not, and for 7 the text did not allow a determination. This is not a criticism of those papers, whose settings differ from ours. It leaves open a practical question: *when does deep RL help link adaptation once the comparison includes rules calibrated on the learner's own cues?*

We study this question in one simulator of a multi-user LTE/NR-like cell with multi-process HARQ, for deadline-constrained and full-buffer traffic. The protocol was built up over 17 versions of the campaign and fixed before the final rounds: disjoint calibration, validation and evaluation traces; common random numbers (CRN) [8]; ten training seeds; nine comparators, including OLLA variants and calibrated rules; a fairness guard; headroom bounds; seed-level inference next to trace-level tests; pre-registered decision rules and wording; and a public registry of every case.

The answer is narrower than most of the literature suggests. It has three parts.

1) *The comparator decides the size of the gain.* For deadline traffic with an HARQ round-trip time (RTT) of 8 TTIs, a residual advantage actor-critic (A2C) policy on a calibrated base rule beats OLLA by 7.49 % in on-time delivery after a 200-TTI warm-up. Against rules re-calibrated by exhaustive search on 500 fresh traces it still wins on evaluation traces, but over the strongest one, a table that combines deadline slack, user class and HARQ-process state, the margin is only +0.66 % (90 % CI [+0.29, +1.03] %). Re-calibration removes 58 % of the margin over the best rule.
2) *The remaining gain is not robust across training seeds.* Only 5 of 10 trained policies beat the combined table; the hierarchical seed×trace 95 % interval is [−0.30, +1.70] %, and a randomly chosen policy beats it on a randomly chosen trace with probability 0.55. The gain is also uneven: for users slower than 10 m/s the margin over the original Context table is negative (−1.71 %).
3) *Where the decision has no structure to exploit, the learner adds nothing.* On full-buffer goodput it is −0.88 % against a single calibrated offset, and at RTT 8

it collapses to its base rule.

A design ablation (NR1) and a pre-registered confirmatory replication on fresh traces and seeds (NR2) show that the oracle class information in the base rule is not needed: the residual policy on a non-oracle base beat the frozen combined table at trace level and robustly across training seeds (+3.35 %, hierarchical 95 % CI [2.22, 4.40] %), although that table uses the true speed class and the policy does not.

Our contributions are:

- **C1, protocol and artefacts.** A pre-registered evaluation protocol for learned LA with calibrated comparators, an information table, CRN, disjoint splits, seed-level inference and multiplicity accounting over a registry of 67 cases and 108 scheme evaluations; code, traces and logs are released.
- **C2, a positive case and its limits.** For deadline traffic under realistic HARQ timing, residual A2C beats all calibrated comparators on evaluation traces, including rules re-calibrated by exhaustive search on 500 fresh traces and a table that combines deadline slack, user class and HARQ-process state; the margin over that combined table is not robust to seed resampling (hierarchical 95 % CI [−0.30, +1.70] %). The trace-level margin over that table is +0.66 %. We recommend the non-oracle residual configuration: in a confirmatory replication (NR2) it beat every calibrated comparator, robustly across seeds. At two neighbouring operating points it beat the OLLA family and the single-cue rules re-calibrated for each point (NR-G10); there the combined table was not in the pre-registered family, so that comparison is established at the M1 point only. The gain is concentrated in faster users; for users below 10 m/s the margin over the original Context table is −1.71 %.
- **C3, where RL does not help.** Under full-buffer goodput the learned policy does not beat a calibrated static offset (−0.88 % and −1.73 % vs. Fixed in M3 and M4).
- **C4, pitfalls and recommendations.** Three simulation and design errors produced large apparent gains in earlier versions of this campaign: a mobility time-scale error, slot stealing by a shared reward, and the timing of the proportional-fair (PF) average under multi-process HARQ. We describe them and derive reporting recommendations.

Section II reviews related work. Sections III–V give the system model, the learning method and the protocol. Section VI reports the results and the pre-registered robustness checks, Section VII the pitfalls, and Section VIII the discussion and limitations. The appendices give hyper-parameters, the full comparator tables and the calibrated tables.

II. Related Work

A. Classical link adaptation and OLLA

Outer-loop link adaptation (OLLA) corrects the reported channel quality indicator (CQI) with an offset that steps down after a NACK and up after an ACK, so that the long-run block-error rate approaches a target [1], [9]. Variants adapt the target [10], the step sizes [11], [12], [13], the offset estimator [14], [15], or the feedback used under URLLC soft-ACK [16]. Ahsen et al. use a rank-dependent multi-offset rule [17]; this is a different design from the Multi-OLLA comparator used here. SALAD varies three OLLA step sizes without learning [18].

B. Learned link adaptation

Bandit formulations choose the modulation and coding scheme (MCS) or the offset directly [19], [2], [20], [21]. Deep reinforcement learning (RL) has been applied with DQN, actor-critic, TD3 and PPO agents [3], [22], [4], [23], [7]. Deadline-aware RL on an OLLA loop is studied by Kela et al. [6], and learning on top of OLLA by Pulliyakode and Kalyanasundaram [24], Pang et al. [22] and Wang et al. (LOLLA) [5]. These works are evaluated in different settings from ours (LOLLA, for example, treats uplink with an immediate CRC outcome and no HARQ feedback delay), and we do not re-implement them. In O-RAN, learning-based control is placed by timescale: xSlice, a PPO agent with a graph convolutional network, runs as an xApp on the near-real-time RIC, whose control loop spans 10 ms to 1 s, and re-partitions resource blocks between slices over the air [25], whereas LOLLA runs as a dApp [26] inside the GPU-accelerated stack and sets the MCS in every slot, with end-to-end control latency below 500 μs at 30 kHz subcarrier spacing [5]. Our question is a different one: given a simulator with multi-process HARQ, how much of a learner's gain over OLLA remains once the comparison includes rules calibrated on the learner's own cues?

Table I summarises eight closely related papers on two points that matter for that question. Across the 23 learning-based papers we screened, 0 compared against a calibrated rule built on the learner's cues, 3 did so partly, 13 did not, and 7 could not be determined. These counts rest on abstracts and the text we could access, and an "unclear" entry is not a negative finding.

C. Evaluation rigour

Doherty et al. show that carefully tuned heuristics matched published RL results on optical-network problems [27]; this is the closest methodological precedent, in another domain. In recommender systems, Ferrari Dacrema et al. could reproduce only a minority of recent neural methods, and most of those were matched or beaten by simple, well-tuned heuristics [28]. Similar concerns about baselines, leakage and reproducibility are raised for learned systems and wireless machine learning in [29], [30], [31]. For RL specifically, variance across seeds and the reporting of intervals are discussed in [32], [33]; Agarwal et al. recommend interval estimates, score distributions over runs and the probability of improvement instead of point estimates [34], and we follow these recommendations. Pre-registration is advocated in [35], [36], and credibility of simulation studies in [37], [38]. We apply these ideas to one concrete question in link adaptation.

TABLE I
EIGHT CLOSELY RELATED LEARNED LINK-ADAPTATION PAPERS. "CALIBRATED RULE": A NON-LEARNING RULE CALIBRATED ON THE LEARNER'S CUES IS A BASELINE. "PARITY": THE LEARNER AND THAT RULE USE COMPARABLE INFORMATION. ENTRIES COME FROM THE ABSTRACT AND ACCESSIBLE TEXT.

Paper	Setting	HARQ model	Calibr. rule	Parity
Saxena [2]	single link, AWGN and fading	first transmission only	no	partial
Ye [3]	LTE/NR simulation	not stated	no	unclear
Kela [6]	5G UL, XR, system-level	system-level HARQ	no	no
Pang [22]	LTE/NR simulation	not stated	no	unclear
You [4]	simulation and USRP/srsRAN testbed	parallel HARQ, ACK delay	partial	no
Wang (LOLLA) [5]	GPU NR stack, UL	immediate CRC, no HARQ delay	no	no
Demirel [7]	system-level, multi-cell	system-level	partial	unclear
Wiesmayr (SALAD) [18]	OAI, single user	real HARQ	partial	yes

III. SYSTEM MODEL

All results come from one MATLAB simulator, evaluated in simulation only. Table II lists the parameters, and each is checked against the code by the number-generation script.

A. Cell and channel

A single macro cell serves $N = 10$ users on a $20\,\mathrm{MHz}$ carrier at $2\,\mathrm{GHz}$ with 100 resource blocks (RBs), $15\,\mathrm{kHz}$ subcarrier spacing and a $1\,\mathrm{ms}$ transmission time interval (TTI). Large-scale fading is log-normal shadowing with $\sigma = 6$ dB and 50 m decorrelation distance; small-scale fading is Jakes. User speeds are drawn from 3–30 m/s, which gives Doppler frequencies of about $20\,\mathrm{Hz}$ to $200\,\mathrm{Hz}$. A two-state Markov process adds interference bursts of 9 dB with mean dwell times of 250 TTIs (calm) and 120 TTIs (burst).

B. CQI feedback and link abstraction

Each user reports a 15-level channel quality indicator (CQI) with a delay drawn from 2–8 TTIs and Gaussian report noise of $\sigma = 1$ dB. Block errors follow an SINR-to-BLER abstraction rather than LDPC link-level tables; this is a limitation (Section VIII). The OLLA reference loop adds an offset Δ to the reported CQI. A NACK lowers Δ by $\Delta_{\downarrow} = 0.5$ dB, and an ACK raises it by $\Delta_{\uparrow} = \Delta_{\downarrow} P_{\mathrm{tgt}}/(1 - P_{\mathrm{tgt}}) \approx 0.056$ dB, for a target $P_{\mathrm{tgt}} = 0.1$, with $|\Delta| \leq 10$ dB. The corrected CQI selects the MCS. We call this corrected CQI the *OLLA reference*. All calibrated rules and the learned action act as CQI-index offsets on top of it.

C. HARQ

Retransmissions use chase combining ($2.5\,\mathrm{dB}$ gain per retransmission, at most 3 retransmissions) and reuse the stored MCS and transport block size. The learner's action therefore affects new transmissions only. We study two timing regimes: *realistic* HARQ with an ACK/NACK round-trip time (RTT) of 8 TTIs and 8 processes per user, and *instantaneous* feedback (RTT 0, one process).

TABLE II
SIMULATION PARAMETERS (CHECKED AGAINST `defaultEnv_v17.m` AND `v17_setup.m`).

Parameter	Value
Users, bandwidth, RBs	10, 20 MHz, 100
Carrier, subcarrier spacing, TTI	2 GHz, 15 kHz, 1 ms
Shadowing	6 dB, 50 m
Speed range (Doppler)	3–30 m/s (20–200 Hz)
Interference burst	9 dB; dwell 250/120 TTI
CQI delay, noise	2–8 TTI, 1 dB
OLLA target, steps, limit	0.1, 0.5/0.056 dB, 10 dB
HARQ	chase, 2.5 dB, ≤ 3 retx; RTT 8/8 proc. or RTT 0/1 proc.
PF averaging	scheduling time, $\alpha = 0.05$
Deadline traffic	12000 bit / 20 TTI, budget 10 TTI
Trace length, burn-in	200, 50 TTI
Evaluation traces	500 per case

D. Scheduler

A proportional-fair (PF) scheduler gives all RBs to one user per TTI. The PF average rate uses an exponential average with $\alpha = 0.05$ and is updated *at scheduling time* with the scheduled rate. Updating it instead when the ACK arrives is a common alternative; under multi-process HARQ it changes conclusions (Section VII), so we state the choice explicitly (Fig. 1b).

E. Traffic and metrics

Deadline traffic: $12\,000\,\mathrm{bit}$ packets arrive every 20 TTIs with a budget of 10 TTIs. The primary metric is the on-time delivery ratio. *Full-buffer traffic:* the primary metric is goodput. Each trace lasts 200 TTIs, of which the first 50 are burn-in, leaving 150 scored TTIs. We also report the Jain fairness index over per-user means and the 5th-percentile user value (edge5). Four cases form the main campaign: M1 (deadline, RTT 8), M2 (deadline, RTT 0), M3 (goodput, RTT 8) and M4 (goodput, RTT 0).

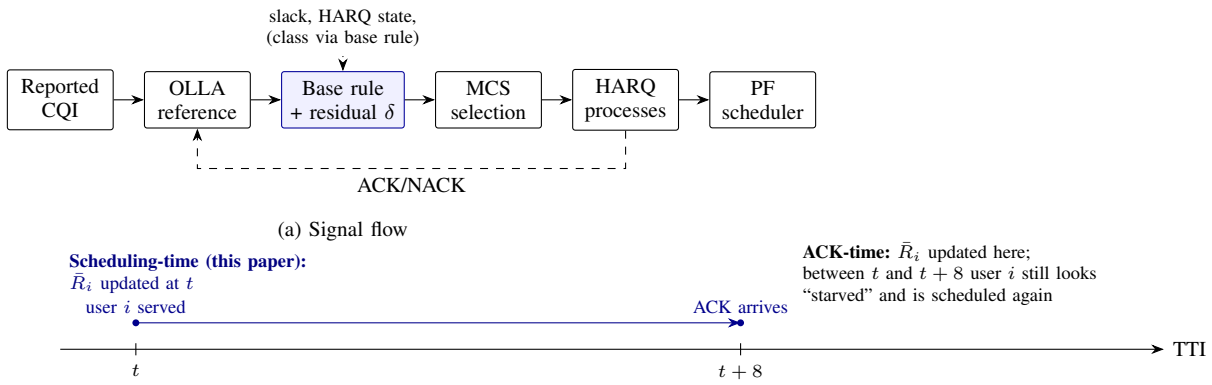
IV. LEARNING FORMULATION AND METHOD

A. Decision problem

At every TTI each scheduled user receives one decision. The effective CQI is $\mathrm{clip}(q_{\mathrm{ref}} + b_i + \delta)$, where q_{ref} is the OLLA reference, b_i is a calibrated base-rule offset and δ is the learned part. In the *offset* formulation $b_i = 0$ and $\delta \in [-4, +2]$. In the *residual* formulation $\delta \in \{-1, 0, +1\}$ and the base rule is the better of Fixed and FixedPU, chosen on calibration traces (it was FixedPU in all four cases). FixedPU uses the true delay and speed class of each user and is therefore a *class oracle* (Section V).

B. Rewards

For deadline traffic the reward is $+1$ for an on-time and -1 for a missed packet, with potential-based shaping $\Phi(s) = -0.5\, r/B$ for r remaining bits of a B-bit packet [39] and



(a) Signal flow

(b) PF average timing at RTT 8

Fig. 1. (a) Decision path: the learner (shaded) acts as a bounded offset on a calibrated base rule applied to the OLLA reference CQI; the dashed arrow is the OLLA loop. (b) The two PF-average definitions on an RTT-8 timeline.

a deviation penalty of $0.02|\delta|$. For goodput the reward is an opportunity cost, the fix for the slot-stealing failure in Section VII.

C. Learner and training

A shared multi-user advantage actor-critic (A2C) [40] uses a 57-128-96-3 actor on three stacked frames of 19 features. It has 20 099 parameters and 19 872 multiply-accumulates per user decision. The logit of $\delta = 0$ starts with a bias of 3. We train for 3000 episodes (deadline) or 800 (goodput) with 10 independent seeds. The *primary scheme* is the per-seed mean: each seed's policy is evaluated and the results are averaged. A deterministic ensemble of the seeds is a secondary scheme. Hyper-parameters are in Appendix A. That is 40 195 floating-point operations per user decision, a hardware-independent cost; wall-clock decision times and where such a policy can run are discussed in Section VIII-D.

V. Evaluation Protocol

A. Rounds, cases and registry

The campaign ran in numbered rounds (v9 to v17). The main round, v17, provides the four cases M1–M4. Each round uses fresh, disjoint seeds and traces: 500 evaluation traces and 10 training seeds for the headline cases. Every case is recorded in a public registry of 67 cases and 108 scheme evaluations (v9 to v17, Aug–Nov 2026). Of these, 7 cases (18 evaluations) were later superseded after an error was found (Section VII), and 49 cases date from v10, when the present protocol began.

B. Comparators and information

Nine comparators are evaluated (seven distinct ones in M1, because the tuned-target OLLA equals the default OLLA and the two bandits coincide there). *Baseline*: open-loop CQI. *OLLA*, *OLLA (tuned target)* and *Multi-OLLA* (a fast/slow blend). *Fixed*: one global offset on the OLLA reference. *FixedPU*: one offset per user class on the OLLA reference, where the class is the true delay class (two) and the true mean-speed class (three); this is a *class oracle*, because the true speed is not observable at the gNB. *Context table*: one offset per remaining-slack bin (three bins). Two contextual bandits (global; per delay class) are trained on the learner's episodes.

Table III states what each scheme observes. The learner's state contains only gNB-observable quantities and adds the slack cue and three HARQ-process features. The residual policy, however, acts on top of FixedPU, so it inherits the class information of that base rule, including the oracle speed class. We therefore describe the calibrated rules as *built on the learner's cues*: the Context table uses the slack cue, FixedPU the class information.

Original and re-calibrated rules. We call the rules calibrated in the main round the *original rules*. A code review before the final comparison showed that their calibration was weaker than "calibrated on the learner's cues" suggests. The Context table came from a two-pass coordinate ascent on 10 traces that did not examine all 343 candidate tables, and the FixedPU table was assembled from per-class optima without ever being evaluated as a whole. A pre-registered re-calibration study removes both limits: every rule is searched again on 500 fresh traces, and a *combined table* that uses slack, class and HARQ-process state together is added (Section VI-B; details in Appendix C). With the combined table the comparison becomes a same-information comparison, which the learner wins on evaluation traces but not robustly across training seeds.

C. Splits, common random numbers and decision rules

Calibration traces (seeds from 60001), validation traces (from 80001) and evaluation traces (from 500001) are disjoint. All schemes see the same evaluation traces with common random numbers [8], so comparisons are paired per trace. Table IV defines the terms used for the comparisons. Figure 2 shows how the pieces fit together.

Seed-level inference uses a hierarchical seed×trace bootstrap [43] (20 000 resamples), a bootstrap over the ten per-seed means and a one-sided sign test over seeds. The two rules only select pre-registered wording; effect sizes and their intervals are reported throughout, so that size and uncertainty can be

TABLE III
INFORMATION USED BY EACH SCHEME. ✓: USED; –: NOT USED; "ORACLE": SIMULATOR GROUND TRUTH, NOT A GNB MEASUREMENT.

Scheme	CQI	ACK hist.	Slack	Delay cls.	Speed cls.	HARQ state	Other	Calibrated on
Baseline (open loop)	✓	–	–	–	–	–	–	–
OLLA	✓	✓	–	–	–	–	–	– (standard)
OLLA, tuned target	✓	✓	–	–	–	–	–	target grid, 10 traces
Multi-OLLA	✓	✓	–	–	–	–	CQI trend, cell NACK average	–
Fixed	✓	✓	–	–	–	–	–	offset grid, 20 traces
FixedPU (class oracle)	✓	✓	–	✓	✓ *oracle*	–	–	per-class grid, 20 traces
Context table	✓	✓	✓(3 bins)	–	–	–	–	coord. ascent, 10 traces
Bandit (global)	✓	✓	–	–	–	–	–	learner's episodes
Bandit (delay class)	✓	✓	–	✓	–	–	–	learner's episodes
A2C, offset action	✓	✓	✓	–	–	✓	ΔCQI, observed SINR, OLLA offset, 3 frames	800 ep. × 10 seeds
Residual A2C (primary)	✓	✓	✓	✓ (base)	✓ *oracle (base)*	✓	as offset A2C	3000 ep. × 10 seeds
Combined table (re-cal.)	✓	✓	✓(3 bins)	✓	✓ *oracle*	✓ (busy)	–	coord. ascent, 500 traces
Residual A2C, Fixed base	✓	✓	✓	–	–	✓	as offset A2C	3000 ep. × 10 seeds

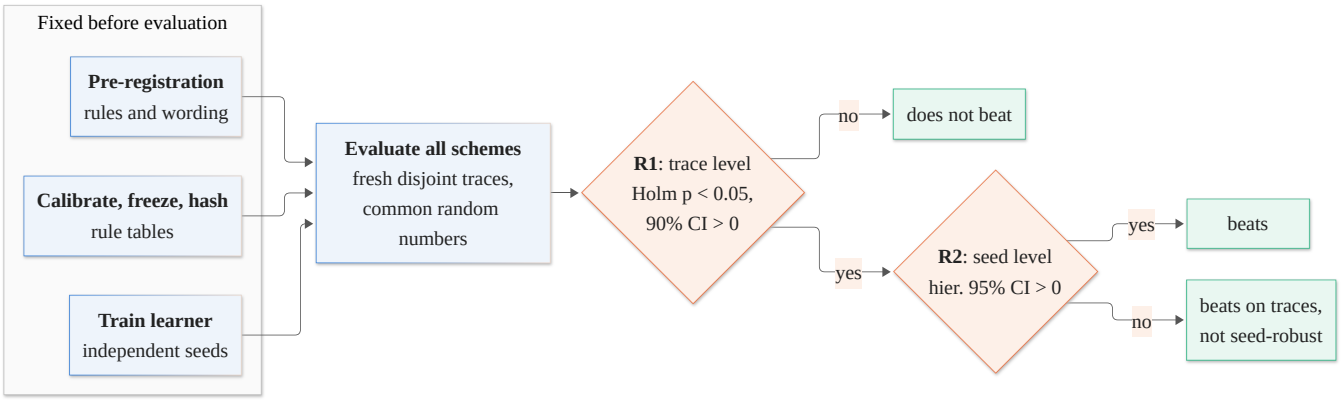


Fig. 2. Evaluation protocol. Everything on the left is fixed before any evaluation trace is drawn: the pre-registration, the calibrated and hashed rule tables, and the trained policies. All schemes are then evaluated on the same fresh traces. The trace-level rule R1 and the seed-level rule R2 (Table IV) only select pre-registered wording; numbers and wording reach the text solely through generated macros.

judged directly rather than through a binary verdict [34]. A fairness guard applies: a Jain-index drop above 0.02 or an edge5 drop above 10 % relative to the best calibrated rule must be reported. Two headroom bounds frame the results: a per-trace hindsight bound and a genie bound.

D. Pre-registration and multiplicity

After the main round, and before any further run, we froze seven short pre-registrations that test the M1 result from different sides; an eighth, the confirmatory replication, was added later by amendment, before it ran (Appendix D). Each text fixes the question, the data, the comparator family, the decision rule and the wording for every outcome, and is hashed. A change after freezing is allowed only before the affected stage starts and is logged as a dated amendment. Calibration tables are hashed before any evaluation trace is touched. The manuscript reads every number from the stages' output files through generated macros, and a correction to a generated verdict is made in the claims ledger and the text, never in the generated file.

From v12 on, a pre-registration note was written before each run, and failed predictions are reported. In v17 the primary predictions were met in all four cases, and two secondary predictions on M1 were not: the 800-episode offset policy was predicted to win and failed Claim B, and the 800-episode residual policy was predicted to fail Claim B and won. Across the registry, 9 of the 60 valid cases show a WIN under the corrected verdict. Multiplicity is handled over the whole registry: the M1 comparison against the best original rule has Holm-adjusted $p = 9.1 \times 10^{-13}$, and $p = 8.8 \times 10^{-10}$ after a Bonferroni factor of 972 (108×9 comparators). The same correction applied to the comparison with the combined table gives $p = 1.00$: that margin passes within its pre-registered family but not across the project. The registry covers the main round; with the 9 pre-registered robustness cases of Appendix D the project has 76 cases and 126 scheme evaluations, at most 1 260 tests, and the two corrected p-values become 1.1×10^{-9} and 1.00. We state plainly that the recipe was chosen during the campaign; M1 is a confirmatory test of that recipe on fresh data.

VI. RESULTS

We first report the main case against the original comparators (Section VI-A). We then report the decisive test,

TABLE IV
TERMS USED IN THE COMPARISONS.

Term	Meaning
Margin	Relative difference $100\,\overline{d}/\overline{x}$ in percent, with d the paired per-trace difference between the learned policy and comparator X, and $\overline{x}$ the comparator's mean. Absolute differences are in percentage points (pp).
Trace-level rule (R1)	"Beats X": Holm-adjusted [41] two-sided Wilcoxon signed-rank $p < 0.05$ [42] over the case's comparator family, *and* the lower bound of the 90 % t-interval of the margin above zero.
Seed-level rule (R2)	A claim that meets R1 but whose hierarchical seed×trace 95 % lower bound against the strongest comparator of the claim is not above zero is reported as "not robust to seed resampling".
Claim A / Claim B	Beats the OLLA family / beats the calibrated rules (Fixed, FixedPU, Context table). Beating all comparators is a *win*.
Primary scheme	The per-seed mean: each of the ten trained policies is evaluated, and the results are averaged.
Original / re-calibrated rules	Rules calibrated in the main round on 10–20 traces / on 500 fresh traces (Section VI-B).
Combined table	One offset per combination of slack, delay class, speed class and busy HARQ processes.
Probability of improvement	Probability that a randomly chosen trained policy beats the comparator on a randomly chosen trace [34].

against rules re-calibrated on a large budget (Section VI-B), and what happens at seed level (Section VI-C). Further checks follow (Section VI-D), then the settings where the learner did not help (Section VI-E) and what the policy actually does (Section VI-F). Table VII in Section VIII collects the main numbers.

A. Main case: deadline traffic with delayed HARQ feedback (M1)

The residual A2C policy delivers 61.26 % of packets on time in M1, against 60.31 % for the Context table, 60.14 % for Fixed and 54.72 % for OLLA (Fig. 3, Table V). Its margin is +11.96 % over OLLA, +1.87 % over Fixed and +1.58 % over the Context table (90 % CI [+1.23, +1.94] %), the strongest of the original rules. In absolute terms the last margin is +0.95 percentage points. The policy beats all nine comparators under the trace-level rule, and it does so without hurting fairness: the Jain index changes by -0.0025 and the 5th-percentile user improves by +1.2 % against the Context table.

How much of the gain over OLLA do simple calibrated rules already capture? A single calibrated offset (Fixed) captures 83 % of the learner's gain over OLLA, and the Context table 85 %. A comparison against OLLA alone would therefore overstate what the learner adds. These original rules were calibrated on only 20 or 10 traces; the next subsection replaces them with rules calibrated on 500 traces.

B. The decisive test: re-calibrated comparators

The original rules had a small calibration budget, and two of their searches were incomplete (Section V). The re-calibration study gives every rule 500 fresh calibration traces, disjoint from the evaluation traces, and searches Fixed, the Context table and the tuned-target OLLA exhaustively. It also adds a stronger rule, the *combined table*: one CQI offset for each combination of deadline slack, delay class, speed class and busy HARQ processes (36 cells, Appendix C). The combined table uses the same cues as the learner and its base rule, including the oracle speed class. All tables were frozen and hashed before any evaluation trace was run. They also show no sign of over-fitting: the combined table scores 0.6088 on the calibration traces and 0.6086 on the evaluation traces.

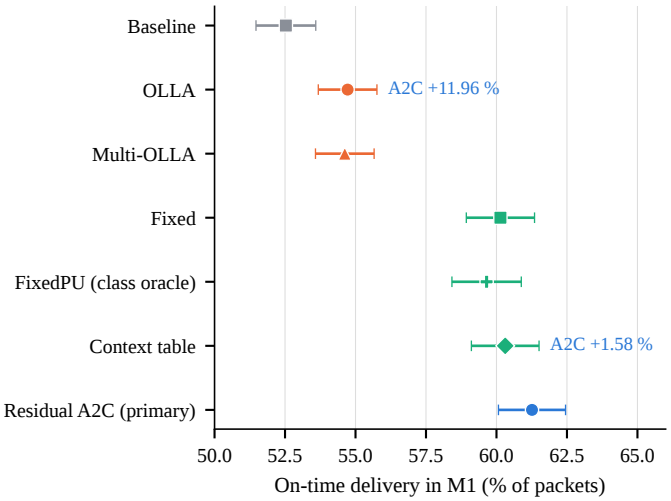


Fig. 3. M1 on-time delivery ratio by scheme (mean and 95 % CI over 500 traces). Blue: residual A2C; orange: OLLA family; green: calibrated rules; gray: baseline. Annotations give the margin of the learned policy.

Table VI and Fig. 6a give the result. The learned policy still beats every re-calibrated rule on the evaluation traces: by +1.43 % over the re-calibrated Context table and +1.87 % over the re-calibrated Fixed offset. *Over the combined table, however, the margin is only +0.66 %* (90 % CI [+0.29, +1.03] %), or +0.40 percentage points. Compared with the +1.58 % over the original Context table, re-calibration and the combined table remove 58 % of the learner's margin over the best rule. Most of what looked like a learning advantage is available to a deterministic table calibrated carefully on the same cues.

Two remarks qualify the combined table. It was expensive to find: the coordinate-ascent search took 24.6 h of simulation, against 2.7 h to train and evaluate the ten learned policies. And one of its 4 search starts stopped at the safety cap while still improving (Appendix C). If anything the table is slightly under-optimised, which favours the learner.

C. Seed-level robustness

Trace-level intervals treat the ten trained policies as fixed. A practitioner, however, trains one policy, so we also ask how the margin varies across training seeds (Fig. 5).

Against the original rules the result is robust. Against the original Context table 9 of 10 seeds are above zero (one-sided sign test $p = 0.0107$), and the hierarchical seed×trace 95 % interval is [+0.62, +2.62] %. Against Fixed it is [+0.89, +2.90] %. The same holds against every re-calibrated single-cue rule (Table VI).

Against the combined table it is not. Only 5 of 10 seeds beat the combined table (sign test $p = 0.623$), and the hierarchical

TABLE V

M1 AGAINST THE ORIGINAL COMPARATORS. MARGIN: PAIRED PERCENTAGE GAIN OF RESIDUAL A2C OVER THE COMPARATOR, WITH THE 90 % CI OVER TRACES AND THE HOLM-ADJUSTED WILCOXON p. "BEATS": THE TRACE-LEVEL RULE IS MET. THE TUNED-TARGET OLLA SELECTED THE DEFAULT TARGET IN M1, AND THE TWO BANDITS COINCIDE. RE-CALIBRATED COMPARATORS ARE IN TABLE VI.

Comparator	On-time (%)	Margin (%)	90 % CI (%)	Holm p	Beats
Baseline	52.53	+16.62	[+15.74, +17.50]	2.8×10^{-76}	yes
OLLA	54.72	+11.96	[+11.21, +12.71]	1.2×10^{-66}	yes
OLLA (tuned target)	54.72	+11.96	[+11.21, +12.71]	1.2×10^{-66}	yes
Multi-OLLA	54.62	+12.16	[+11.40, +12.92]	3.1×10^{-67}	yes
Bandit (global)	41.80	+46.55	[+45.22, +47.88]	1.3×10^{-82}	yes
Bandit (delay class)	41.80	+46.55	[+45.22, +47.88]	1.3×10^{-82}	yes
Fixed	60.14	+1.87	[+1.52, +2.23]	1.3×10^{-16}	yes
FixedPU (class oracle)	59.65	+2.70	[+2.36, +3.03]	1.3×10^{-31}	yes
Context table	60.31	+1.58	[+1.23, +1.94]	9.1×10^{-13}	yes

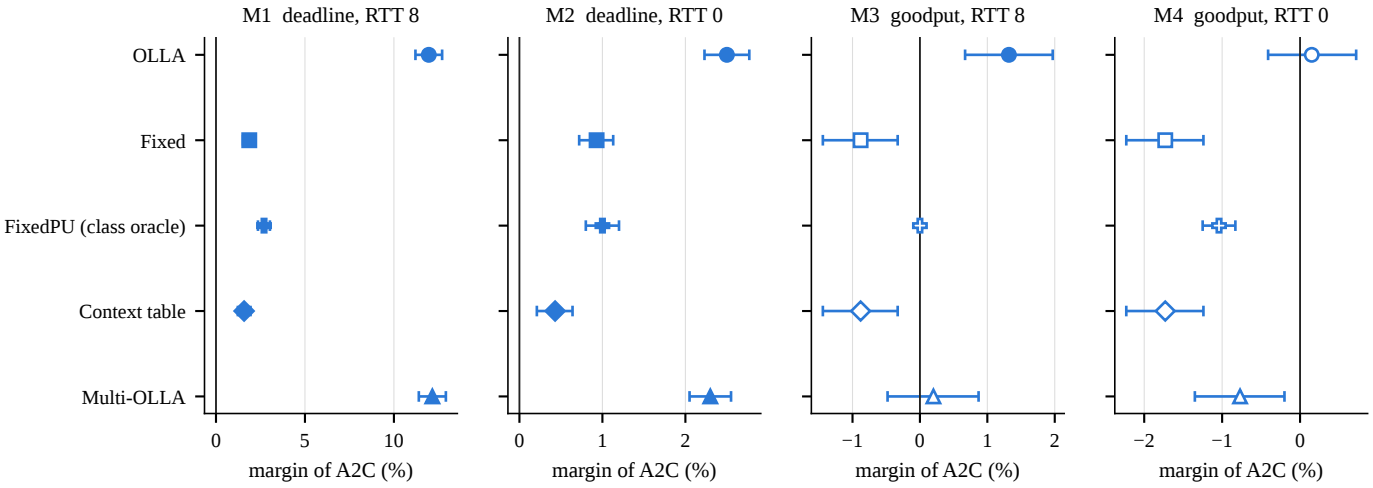


Fig. 4. Margin of residual A2C over each original comparator in the four cases (90 % CI over traces). Filled markers: the trace-level rule is met; open markers: it is not. The two bandits are omitted for scale (see Table V).

interval is [−0.30, +1.70] %. Under the pre-registered seed-level rule the margin is therefore *not robust to seed resampling*. The probability that a randomly chosen trained policy beats the table on a randomly chosen trace [34] is 0.55 (95 % CI [0.49, 0.61]), close to a coin flip. The trace-level result also fails a Bonferroni correction over all tests in the project ($p = 1.00$, Section V). The automatic summary of the re-calibration stage had marked the result as seed-robust, because it applied the seed-level rule to the re-calibrated Context table only; the rule requires the strongest comparator, and we report the corrected verdict.

The hierarchical intervals are about 2.3 times wider than trace-only ones, which is the price of counting seed variance. A deterministic ensemble of the ten seeds (a secondary scheme) reaches 0.6306 on time, which shows how much of the spread an ensemble removes; it was not part of the pre-registered decision.

D. Further checks

1) Does OLLA just need time to converge?: OLLA starts each trace cold, and only 150 TTIs are scored. In the warm-up check all loops run for 200 TTIs before scoring. The margin over OLLA falls from +11.96 % to +7.49 % (90 % CI [+6.79, +8.19] %), so part of the gap to OLLA was its convergence time. The learner itself barely changes (0.6126 cold, 0.6116 warm). Against the calibrated rules the warm-up changes little:

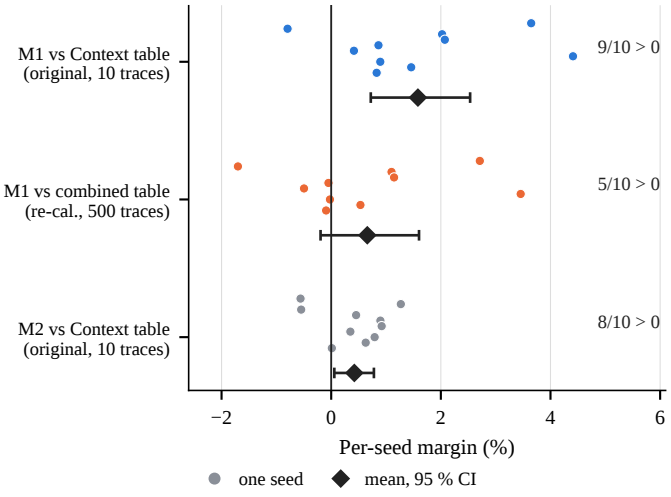


Fig. 5. Per-seed margins (one dot per training seed) with the mean and its 95 % bootstrap interval over seed means (diamond). Top: M1 against the original Context table. Middle: M1 against the combined table. Bottom: M2 against the original Context table. The count of seeds above zero is given on the right.

TABLE VI

THE LEARNED POLICY (M1) AGAINST COMPARATORS RE-CALIBRATED ON 500 FRESH TRACES, ON THE 500 M1 EVALUATION TRACES. TRACE LEVEL: MARGIN, 90 % CI AND HOLM-ADJUSTED p OVER THESE 10 COMPARATORS. SEED LEVEL: SEEDS WITH A POSITIVE MARGIN, HIERARCHICAL SEED×TRACE 95 % CI, AND WHETHER THE SEED-LEVEL RULE IS MET (LOWER BOUND ABOVE ZERO).

	Trace level				Seed level		
Comparator	Margin (%)	90 % CI (%)	Holm p	Beats	Seeds > 0	Hier. 95 % CI (%)	Robust
Baseline	+16.62	[+15.74, +17.50]	3.2×10^{-76}	yes	–	–	–
OLLA	+11.96	[+11.21, +12.71]	1.5×10^{-66}	yes	–	–	–
OLLA (re-tuned)	+8.40	[+7.74, +9.07]	6.0×10^{-55}	yes	10/10	[+7.15, +9.66]	yes
Multi-OLLA	+12.16	[+11.40, +12.92]	3.6×10^{-67}	yes	–	–	–
Fixed (re-cal.)	+1.87	[+1.52, +2.23]	2.7×10^{-16}	yes	9/10	[+0.91, +2.92]	yes
FixedPU (re-cal., class oracle)	+1.73	[+1.31, +2.14]	2.8×10^{-12}	yes	9/10	[+0.72, +2.80]	yes
Context table (re-cal.)	+1.43	[+1.08, +1.77]	6.1×10^{-11}	yes	9/10	[+0.47, +2.46]	yes
Bandit (global)	+46.55	[+45.22, +47.88]	1.5×10^{-82}	yes	–	–	–
Bandit (delay class)	+46.55	[+45.22, +47.88]	1.5×10^{-82}	yes	–	–	–
Combined table	+0.66	[+0.29, +1.03]	1.0×10^{-3}	yes	5/10	[−0.30, +1.70]	**no**

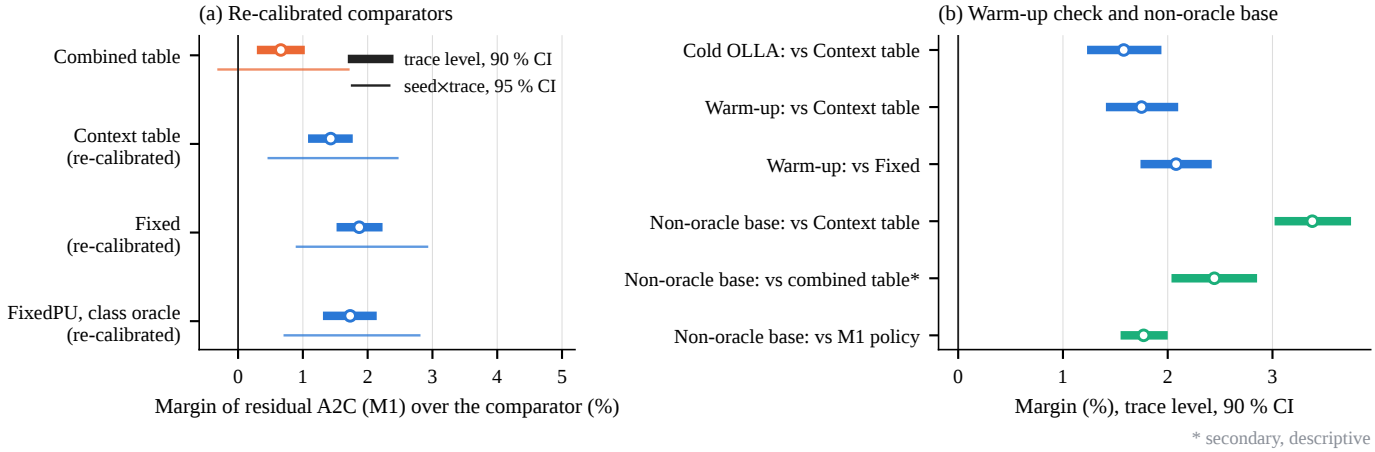


Fig. 6. (a) Margin of the learned policy over the re-calibrated comparators; thick bars: trace-level 90 % CI; thin bars: hierarchical seed×trace 95 % CI. Orange: the seed-level interval includes zero. (b) Warm-up check (OLLA warmed up for 200 TTIs) and non-oracle base ablation; trace-level 90 % CIs. The ablation has no seed-level interval because per-seed values were not stored.

+1.75 % over the Context table and +2.08 % over Fixed. As pre-registered, the abstract quotes the smaller, warm-up gain over OLLA.

2) A non-oracle configuration (NR1, NR2): The M1 policy acts on FixedPU, whose offsets use the true speed class. The non-oracle base ablation (NR1) retrains the same recipe on the Fixed base (one global offset of −2 CQI steps, the same value as the re-calibrated Fixed rule), with fresh training seeds. The learner's own state is gNB-observable, so this configuration uses no oracle information at all. It delivers 0.6235 on time, beats all 9 original comparators (+3.38 % over the Context table) and is +1.77 % above the M1 policy ([+1.55, +2.00] %; Fig. 6b). The gain does not require class-oracle information in the base rule. A secondary comparison on the same traces puts it +2.44 % above the combined table ([+2.04, +2.85] %).

Because NR1 reused M1's evaluation traces and did not store per-seed values, a pre-registered confirmatory replication (NR2) was conducted with 500 fresh evaluation traces, 10 fresh training seeds, and the frozen combined table Context-joint-500 as primary comparator. On fresh evaluation traces and fresh training seeds, the residual policy on a non-oracle base beat the frozen combined table at trace level and robustly across training seeds (3.35 %, hierarchical 95 % CI [2.22, 4.40] %). On the fresh traces, the non-oracle policy also exceeded the M1 oracle-base policies by 2.58 %.

3) Does it hold at other operating points?: The operating-point check tests two neighbouring points: a higher load at the M1 deadline (G2), and a lower load with a looser deadline (G3). The comparators are the OLLA family (OLLA, OLLA with a tuned target, Multi-OLLA), the bandits and the single-cue calibrated rules (Fixed, FixedPU and the Context table), re-calibrated for each point by exhaustive search on that point's own calibration traces. The combined table is not in the pre-registered family at these points: it was dropped for compute reasons before any run (Amendment 2), so "beats the combined table" remains a statement about the M1 point. With the M1 recipe, whose base-rule selector chose the class-oracle FixedPU at both points, the check was negative: The result was observed at the M1 operating point only; at the two tested neighbouring points it was not reproduced. Two further grid corners were not tested.

We then ran the same check with the recommended non-oracle recipe (NR-G10, pre-registered and frozen before the run; Fixed base with the point's calibrated static offset, 5 fresh training seeds per point, the same 300 evaluation traces and common random numbers as above). With the non-oracle

residual recipe, the gain held at both tested neighbouring operating points (a higher load at the M1 deadline, and a lower load with a looser deadline): residual A2C on a calibrated static offset beat the OLLA family and the single-cue calibrated rules re-calibrated for each cell (margins +1.73 % and +1.02 % vs the best rule). Two further grid corners were not tested. Both margins also pass the seed-level rule against the best rule: hierarchical 95 % CIs [0.90, 2.73] % in G2 and [0.63, 1.42] % in G3; with 5 seeds this is indicative. In G3 the learned policy reaches an on-time ratio of 0.91, close to the ceiling, so the room for any scheme is small there and the margin is small in absolute terms.

Descriptive, not pre-registered. The G10 stage also calibrated and evaluated a per-cell combined table, with a lighter search than at M1 (200 calibration traces per point, 4 starts, instead of 500 traces), although it was outside the pre-registered family. Because NR-G10 reused G10's evaluation traces and common random numbers (the nine shared comparators are bit-identical in both runs), the two can be paired per trace. Against this table the non-oracle policy is +1.04 % in G2 (90 % CI [0.49, 1.59] %, hierarchical 95 % CI [0.14, 2.08] %) and +1.17 % in G3 ([0.87, 1.47] %; [0.75, 1.60] %), with 5 and 5 of 5 seeds above zero. No decision rule is attached to this comparison and it does not enter the verdict.

E. Where the learner did not help

Instantaneous feedback (M2). Without HARQ delay the margin over the Context table shrinks to +0.43 % (90 % lower bound +0.21 %), and the hierarchical interval [−0.03, +0.87] % includes zero; 8 of ten seeds are above zero (Fig. 5, bottom). We therefore describe M2 as a replication only. Appendix B gives the full M2–M4 tables.

Full-buffer goodput (M3, M4). Here the learner does not beat a single calibrated offset: −0.88 % (M3) and −1.73 % (M4) against Fixed. At RTT 8 it is bit-identical to its base rule in 5000 of 5000 seed×trace cells: on all 379 380 scheduled new transmissions it chose $\delta = 0$, so it learned to leave the base rule alone. At RTT 0 it did act (2985 of 5000 cells are identical) and did worse than its base rule.

How much room was there? The per-trace hindsight bound, the best single offset per trace, lies +3.4 % above Fixed in M1 (+2.0 % in M2), and a genie bound +37.8 % (Fig. 7a). A policy that adapts within a trace can exceed the hindsight bound. Against these bounds the learner's +1.87 % over Fixed is a modest share of the room. Training length matters: at 800 episodes the margin over Fixed is +0.62 %, at 3000 episodes +1.87 % (Fig. 7b), and the shorter variant has only 6 of ten seeds above the Context table.

F. What the policy does

1) Actions: On 587 016 scheduled new transmissions in M1 the policy raises the CQI by one step ($\delta = +1$) in 54.6 % of decisions, keeps the base rule in 26.4 % and lowers it in 19.0 % (Fig. 8b). The ten policies differ widely: the share of upward steps ranges from 11 to 78 % across seeds, although their on-time ratios are close. Three patterns hold in every seed. Upward steps are more frequent with one or two busy HARQ processes than with none (84.7 % against 37.0 %), when at most 0.6 of the deadline budget remains (89.6 % against 48.4 %), and during interference bursts (69.6 % against 48.7 %). This is consistent with a policy that accepts a higher block-error risk for a larger transport block when retransmissions are pending or the deadline is close. We have not tested this mechanism.

2) Users at different speeds: Splitting M1 by each user's mean speed and comparing with the original Context table shows an uneven gain (Fig. 8a). Users between 10 and 20 m/s gain +2.82 % (hierarchical 95 % CI [+1.80, +3.89] %) and faster users +2.25 %. For users slower than 10 m/s, 21 % of user-traces, the margin is *negative*: −1.71 % ([−3.32, −0.15] %). The policy's actions hardly depend on speed (51.9, 54.3 and 56.7 % upward steps from the slowest to the fastest class). This is consistent with a largely speed-independent correction that helps when the channel ages quickly between report and use and over-shoots when it ages slowly. This breakdown was not repeated against the re-calibrated rules.

VII. Pitfalls and Recommendations

Three simulation or design errors changed conclusions during the campaign. We describe them because each can recur in other link-adaptation studies.

A. Mobility time scale

Early versions (up to v9) moved users v metres per TTI instead of $v\,T_{\mathrm{tti}}$. Channel dynamics were then about a thousand times too fast. Under this error the learner appeared to beat OLLA; after the correction, the v9 registry shows 0 wins in 18 cases, and the A2C policy was −4.08 % against OLLA on goodput.

B. Slot stealing

A per-user goodput reward let a shared agent lower one user's rate to benefit others' slots. The learned policy was bimodal (58 % of actions at -3 and 35.8 % at $+2$), and goodput was 1.38 Mbps against 2.92 for OLLA. An opportunity-cost reward removed the effect (2.9191 against 2.9233 Mbps, a -0.14 % difference that is not significant).

C. PF-average timing under multi-process HARQ

If the PF average is updated with ACKed bits at $t + \mathrm{RTT}$, a just-served user looks starved for RTT TTIs and is scheduled again. In a six-trace diagnostic, Baseline goodput at RTT 8 rose from 3.466 to 4.837 Mbps when going from one to eight processes (+39.6 %) while the Jain index fell from 0.503 to 0.420. Re-running all 13 schemes with the other PF definition changed goodput by 10–26 %, but that comparison is confounded by traces, drain and useful-bit accounting, so it is quoted only as a range. A 10.71 % engine discontinuity had the same cause. The sign of a simple comparison depends on the definition: open-loop Baseline was +4.5 % above OLLA under delivered-throughput PF (v9) but OLLA is +7.4 % above Baseline under scheduling-time PF in M3 (and +6.9 % in M4). A controlled demonstration on 100 fresh traces (Fig. 9) isolates

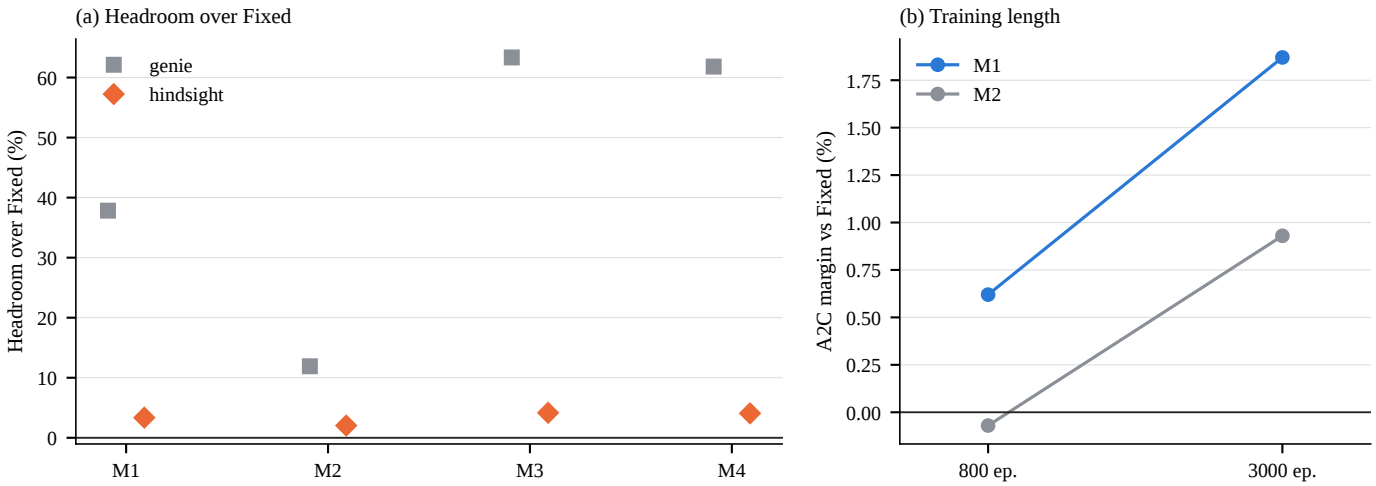


Fig. 7. (a) Headroom over Fixed in the four cases: genie and per-trace hindsight bounds. (b) Margin of residual A2C over Fixed after 800 and 3000 training episodes.

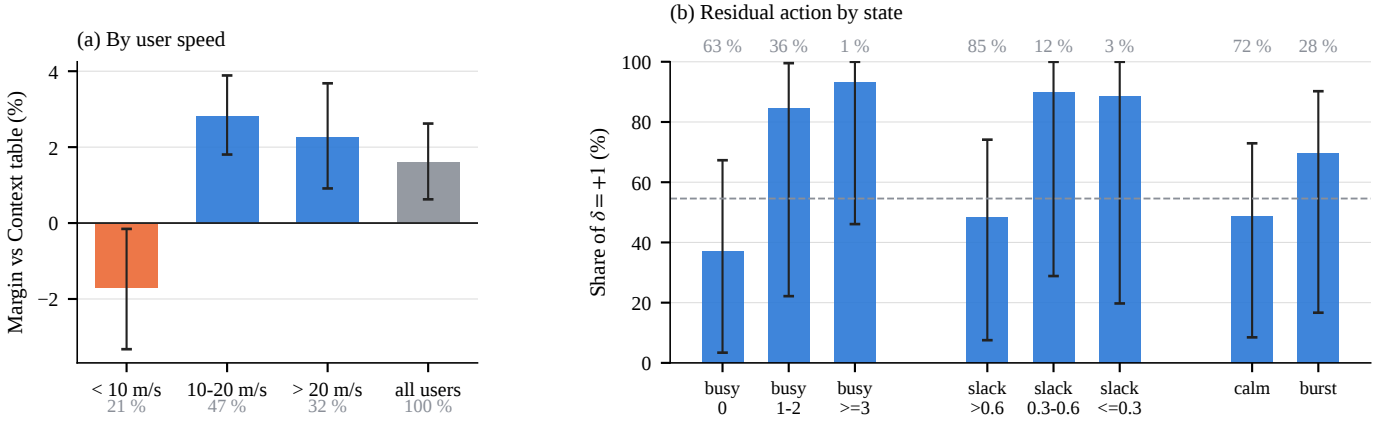


Fig. 8. What the policy does in M1. (a) Margin over the original Context table by user mean speed, with hierarchical 95 % CIs; gray labels give the share of user-traces. The all-user value pools user-traces, so it differs slightly from Table V. (b) Share of upward steps ($\delta = +1$) on scheduled new transmissions by state, pooled over seeds (bars), with the minimum and maximum over the ten seeds (whiskers); gray labels give the share of decisions in each bin; dashed line: all decisions. Slack is the remaining fraction of the deadline budget.

the effect. Under ACK-time averaging, going from one to eight processes at RTT 8 raises Baseline goodput by a factor of 1.447 (2.92 to 4.23 Mbps) and lowers the Jain index from 0.534 to 0.436. Under scheduling-time averaging the factor is 0.976 and the Jain index stays at 0.498 (from 0.494). OLLA and Fixed behave the same way (factors 1.433 and 1.420 under ACK-time averaging, 0.976 and 1.007 under scheduling-time averaging). The pre-registered expectation held for all three schemes: in this diagnostic, the timing of the PF average, not HARQ itself, accounts for the apparent gain.

D. Withdrawn and superseded results

Results from v7 and v8 are withdrawn. A v10 exploratory win (E4c) used overlapping traces. The v12 pre-registered replication failed Claim B and the v13 X1 prediction failed. The v15 and v16 RTT>0 cases (7 cases, 18 evaluations) are superseded because they used the ACK-time PF average. The registry in the supplement lists every case.

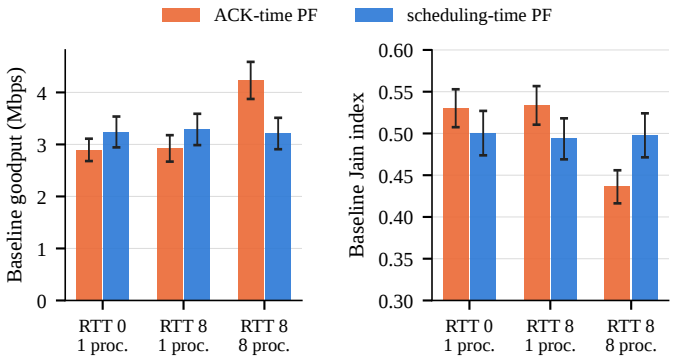


Fig. 9. PF-timing demonstration: Baseline goodput and Jain index under the two PF-average definitions, for RTT 0 with one process and RTT 8 with one or eight processes (mean and 95 % CI over 100 traces).

E. Recommendations

1) Compare against rules calibrated on the learner's cues, state the calibration budget, and give an information

table such as Table III.
2) Use common random numbers, disjoint calibration, validation and evaluation traces, and a public case registry.
3) Report seed-level intervals next to trace-level ones.
4) Report results by mobility regime, HARQ timing and observation set. The observation-set ablation in LOLLA (Table V(a) of [5], 254, 185 and 171 Mbps) shows how much a result can depend on what the learner sees.
5) State the PF-averaging definition and report the complexity of the learner.

VIII. Discussion and Limitations

A. When does RL help link adaptation?

In this study RL helped only where the decision had structure that a static offset cannot use. On full-buffer goodput (M3, M4) every useful input is already summarised by a calibrated offset per user class, and the residual policy learned to leave its base rule alone at RTT 8 and did worse than it at RTT 0. With deadline traffic and delayed HARQ feedback (M1) the policy found something to add. Its corrections are most frequent when HARQ processes are busy, when little of the deadline budget remains and during interference bursts (Section VI-F). This is consistent with the policy trading block-error risk against the chance of finishing a packet in time, a trade-off that a table indexed by a few bins approximates only coarsely. We did not test this mechanism.

Three qualifications bound the positive result. First, the margin over the strongest calibrated rule is small: +0.66 %, or +0.40 percentage points of on-time delivery. Second, it is not robust across training seeds: only 5 of 10 seeds beat the combined table, and the hierarchical interval $[-0.30, +1.70]$ % includes zero. The ten policies also act very differently (the share of upward corrections ranges from 11 to 78 % across seeds). A practitioner who trains a single policy has roughly even odds of beating the combined table: the probability of improvement is 0.55 ([0.49, 0.61]). Third, the gain is not uniform across users. Against the original Context table, users above 10 m/s gain (+2.82 % and +2.25 %), while users below it have a margin of -1.71 % (hierarchical 95 % CI $[-3.32, -0.15]$ %), and these slow users make up 21 % of user-traces. The policy's actions barely depend on speed, so a speed-aware residual or a speed-gated deployment might remove the loss; we have not tried either.

Table VII collects where the learned policy helped in this study, and by how much.

B. Calibrate first

The largest effect in this study is not the learner but the comparator. Against OLLA started cold the learner wins by +11.96 %; a warm-up of 200 TTIs reduces this to +7.49 %. One calibrated offset (Fixed) captures 83 % of the learner's gain over OLLA, and the original Context table captures 85 %. Exhaustive re-calibration on 500 traces and a combined table remove a further 58 % of the remaining margin. Each step changes the conclusion a reader would draw. The steps are not all cheap. Calibrating the single-cue rules on 500 traces took 2.9 h of simulation, but the coordinate-ascent search for the combined table took 24.6 h, against 2.7 h to train and evaluate the ten M1 policies. A fair comparison therefore needs a comparable tuning budget for the rules, and that budget should be reported, as for the learner [28]. We therefore read the M1 result as: *calibrate first*; at the studied point the learned residual then adds a fraction of a percentage point on average, with a seed-dependent sign against the best table.

The calibration study also says something about the base rule. On the evaluation traces the re-calibrated non-oracle Fixed offset scores -0.13 % above the re-calibrated class-oracle FixedPU table, although FixedPU was the base rule selected for M1 on 20 calibration traces. A base rule picked on a small calibration set can thus be the weaker one even when it has more information. This is consistent with the non-oracle base configuration (NR1, NR2). In the pre-registered confirmatory replication (NR2), residual A2C on the non-oracle Fixed base beat the frozen combined table by 3.35 % (90 % CI [2.93, 3.76] %, hierarchical 95 % CI [2.22, 4.40] %), with 9 of 10 seeds positive, despite the combined table observing the true speed class while the learned policy does not. We therefore recommend the non-oracle residual configuration: a non-oracle base calibrated on sufficient traces plus a learned residual.

C. Cost

The actor has 20 099 parameters and needs 19 872 multiply-accumulates (40 195 floating-point operations) per user decision. This count does not depend on hardware and is the cost figure we rely on; wall-clock time is discussed with the placement below. Training is not negligible (3000 episodes per seed, ten seeds), but in this study the search for the strongest comparator cost more simulation time than training the learner (Section VIII-B).

D. Where would it run?

A residual link-adaptation policy acts once per scheduled user and slot: its correction must be ready before that slot's grant is built. An NR slot lasts 1 ms at 15 kHz subcarrier spacing and halves with each numerology step, to 0.5, 0.25 and 0.125 ms at 30, 60 and 120 kHz [44]. The near-real-time RIC closes its loop in 10 ms to 1 s, which suits slicing decisions such as those of xSlice [25] but not a per-slot choice of MCS. Per-TTI link adaptation therefore belongs in the DU, in the MAC scheduler or in a dApp next to it [26], which is where LOLLA runs [5], and not in a near-real-time xApp. Whether the policy fits the slot there depends on the platform, the number of users per slot and the tail of the decision-time distribution, not on the median alone. A first CPU benchmark (A3b) had tails dominated by timer and operating-system noise; the follow-up A3c repeats it single-threaded at high process priority with a measured timer floor. On one thread of a laptop CPU (Intel Core i5-8350U, Windows 10, Python/NumPy), with 10 users per slot, the median per-TTI decision time of residual A2C is 180.4 μs, against 12.3, 31.9 and 33.0 μs for Fixed, the combined table and OLLA (Fig. 10). The policy time includes building the state from the simulator's observations, the forward pass and the action choice; all four schemes run in the same single-threaded

TABLE VII
SUMMARY: WHERE THE RESIDUAL A2C POLICY HELPED. MARGINS ARE TRACE-LEVEL; THE LAST COLUMN GIVES THE HIERARCHICAL SEED×TRACE 95 % CI AGAINST THE SAME COMPARATOR WHERE IT WAS COMPUTED (ROBUST IF THE LOWER BOUND IS ABOVE ZERO). †AGAINST THE RE-TUNED OLLA OF THE RE-CALIBRATION STUDY.

Setting	Comparator	Margin (%)	Beats	Seed level: hier. 95 % CI (%)
M1: deadline, RTT 8	OLLA, cold start / after warm-up	+11.96 / +7.49	yes / yes	[+7.15, +9.66]†
M1	best original rule (Context table, 10 traces)	+1.58	yes	[+0.62, +2.62]
M1	best re-calibrated single-cue rule (Context table)	+1.43	yes	[+0.47, +2.46]
M1	**combined table (re-calibrated)**	+0.66	yes (not after project-wide correction)	**[−0.30, +1.70]**
M1, users < 10 m/s	original Context table	−1.71	–	[−3.32, −0.15]
M1, non-oracle base	combined table	+3.35	yes	[2.22, 4.40]
G2 / G3, non-oracle base	best re-calibrated single-cue rule (Context table)	+1.73 / +1.02	yes / yes	[0.90, 2.73] / [0.63, 1.42]
M2: deadline, RTT 0	best original rule	+0.43	yes	[−0.03, +0.87]
M3: goodput, RTT 8	Fixed	−0.88	no	–
M4: goodput, RTT 0	Fixed	−1.73	no	–

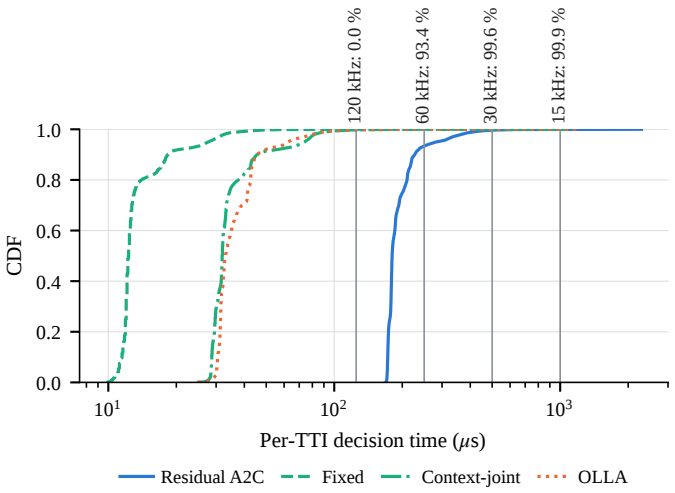


Fig. 10. Per-TTI decision time for 10 users on one thread of a laptop CPU (Intel Core i5-8350U, Windows 10, Python/NumPy): empirical CDF for residual A2C, Fixed, the combined table and OLLA. Vertical lines: slot length at 15, 30, 60 and 120 kHz subcarrier spacing [44]; labels give the share of residual-A2C TTIs under each line. Medians and noise-floor-corrected tails are given in the text; the residual-A2C time includes state construction, but no data collection from a real MAC and no transport, and is not a real-time feasibility result.

Python/NumPy runtime. After subtracting the timer noise floor (the p99.9 of an empty timed loop, 0.5 μs), its p99.9 is 632.1 μs; operating-system scheduling jitter is not removed by this correction. Of the TTIs, 99.9, 99.6, 93.4 and 0.0 % finish within the slot at 15, 30, 60 and 120 kHz. These times come from one general-purpose platform outside a real MAC: an optimised DU implementation would be faster, while data collection from the MAC, transport to a dApp and contention with other DU processes would add time. They bound nothing about a deployment and do not establish real-time feasibility, in particular not at 120 kHz, where no measured TTI met the slot.

E. Limitations

1) *Simulator.* One macro cell, $N = 10$ users and one user scheduled per TTI. There is no multi-cell interference coordination, carrier aggregation or frequency-selective scheduling. Block errors use an SINR-to-BLER abstraction rather than NR LDPC link-level tables, and there is no cross-validation in ns-3 or Sionna. All results come from one MATLAB code base.
2) *No over-the-air validation.* All results are simulated. This is deliberate for the questions asked here: the pre-registered comparisons pair every scheme on the same traces with common random numbers, re-evaluate stored policies on identical channels, and replicate on fresh traces and seeds. Over the air, two schemes never see the same channel, interference or HARQ outcomes, so these paired comparisons and their seed-level intervals cannot be reproduced there. Over-the-air results, as in xSlice [25] or LOLLA [5], answer a different question (does it work on real hardware) and remain future work for this policy. The natural route is a port of the residual policy and its comparators into the downlink scheduler of OpenAirInterface (`openair2/LAYER2/NR_MAC_gNB`) [45], as MAC code or as a dApp [26].
3) *Seed robustness.* The margin of M1 over the combined table is not robust to seed resampling (hierarchical interval includes zero). The recommended non-oracle configuration (NR2) passed the seed-level rule with 9 of 10 seeds positive and hierarchical 95 % CI [2.22, 4.40] %.
4) *Heterogeneous gain.* For users below 10 m/s the margin against the original Context table is −1.71 %. The speed breakdown was not repeated against the re-calibrated rules.
5) *Operating point.* With the M1 recipe: The result was observed at the M1 operating point only; at the two tested neighbouring points it was not reproduced. Two further grid corners were not tested. With the non-oracle residual recipe, the gain held at both tested neighbouring operating points (a higher load at the M1 deadline, and a lower load with a looser deadline): residual A2C on a calibrated static offset beat the OLLA family and the single-cue calibrated rules re-calibrated for each cell

(margins +1.73 % and +1.02 % vs the best rule). Two further grid corners were not tested. At the neighbouring points the strongest pre-registered comparator is a single-cue rule; the combined table was pre-registered at the M1 point only (a descriptive comparison with a per-cell combined table is in Section VI-D).

6) *Oracle information.* The M1 base rule FixedPU uses the true speed class, which the gNB does not observe, and the combined table uses it too. The recommended configuration (NR1, NR2) uses no oracle information; the comparator it beats (the combined table) does. The recommendation was chosen after seeing NR1 and G6 results, but NR2 provides an independent confirmatory test on fresh traces and seeds.
7) *Scoring window.* Traces score 150 TTIs after a burn-in. OLLA starts cold unless warmed up (Section VI-D).
8) *Selection over versions.* The residual recipe was chosen over 17 versions of the campaign. M1 is a confirmatory test of that recipe on fresh data, and multiplicity is accounted for over the registry, but selection effects cannot be fully excluded. The released registry covers the main round (67 cases); the 9 robustness cases are counted in the project-wide correction (Section V) and listed in Appendix D.
9) *Definitions.* Conclusions about multi-process HARQ depend on the PF-averaging definition (Section VII). We use the scheduling-time definition throughout.
10) *Learned baselines.* No learned LA scheme from the literature (for example LOLLA or SALAD) was re-implemented in our simulator.

IX. Conclusion

We asked when deep RL helps downlink link adaptation once it is compared with rules calibrated on the learner's own cues. In one simulator with multi-process HARQ, the answer depends on three conditions. The traffic must give the decision structure that a static offset cannot use: on full-buffer goodput the learner collapsed to its base rule or did worse. The comparator decides the size of the gain: against OLLA it is large, against rules calibrated on the learner's cues it is small. And the claim must survive seed variance: against a carefully calibrated table combining the same cues, the residual policy won on evaluation traces but not robustly across training seeds, and it lost for slow users. Most of the apparent advantage over standard baselines came from comparator calibration. We recommend that learned link-adaptation studies report calibrated comparators with an information table, disjoint splits with common random numbers, seed-level intervals and results by mobility regime. The most promising configuration found here, a learned residual on a non-oracle calibrated offset, passed a pre-registered replication on fresh traces and seeds against the combined table at the M1 point. At two neighbouring operating points it beat the single-cue rules re-calibrated for each point; whether it also beats a fully calibrated combined table there, and at the two untested grid corners, is open.

Code, data and pre-registration availability

[Draft placeholder] Release DOI. The release will contain the simulator, the trace and CRN generators with all seeds, the per-seed and per-trace results of every scheme (so that any statistic in this paper can be recomputed), the calibrated tables with their hashes, the case registry, the analysis scripts, and the frozen pre-registration texts with their SHA-256 hashes and freeze commits.

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[Draft placeholder] Funding, and the AI-assistance disclosure in the wording required by the target journal (see `submission_checklist.md`).

Appendix A
Training details

Table VIII lists the learner's hyper-parameters. They are read from the simulator code by the number-generation script. The same values were used in all cases and checks; the re-calibration study and the warm-up check re-use the M1 policies.

TABLE VIII
Hyper-parameters of the shared multi-user A2C (all cases).

Item	Value
Actor network	57-128-96-3 (residual)
Parameters, MACs per decision	20 099, 19 872
Actor / critic learning rate	3×10^{-4} / 1×10^{-3}
Discount factor γ	0.95
n-step return	10
Entropy weight (start $\rightarrow$ end)	0.03 $\rightarrow$ 0.001
Validation	every 50 episodes on 10 traces
Initial logit bias on $\delta = 0$	3
Deviation penalty	$0.02\,\lvert\delta\rvert$
Shaping constant c	0.5
Episodes (deadline / goodput)	3000 / 800
Training seeds per case	10

Appendix B
Comparator tables for M2–M4

Tables IX–XI give the margin of the primary residual A2C scheme over each original comparator, in the format of Table V. On-time ratio is the metric in M2, goodput in M3 and M4.

TABLE IX
M2 (deadline, RTT 0): residual A2C vs. each comparator.

Comparator	Margin	90 % CI	Holm p	Beats
Baseline	+4.73	[+4.40, +5.06]	2.3×10^{-63}	yes
OLLA	+2.50	[+2.23, +2.77]	8.6×10^{-39}	yes
OLLA (tuned target)	+2.50	[+2.23, +2.77]	8.6×10^{-39}	yes
Multi-OLLA	+2.30	[+2.05, +2.55]	8.6×10^{-39}	yes
Fixed	+0.93	[+0.72, +1.13]	1.1×10^{-10}	yes
FixedPU (class oracle)	+1.00	[+0.80, +1.20]	9.4×10^{-15}	yes
Context table	+0.43	[+0.21, +0.64]	1.7×10^{-2}	yes
Bandit (global)	+12.57	[+12.09, +13.06]	3.8×10^{-82}	yes
Bandit (delay class)	+12.57	[+12.09, +13.06]	3.8×10^{-82}	yes

TABLE X
M3 (GOODPUT, RTT 8): RESIDUAL A2C VS. EACH COMPARATOR.

Comparator	Margin	90 % CI	Holm p	Beats
Baseline	+8.81	[+7.93, +9.70]	3.8×10^{-44}	yes
OLLA	+1.32	[+0.67, +1.97]	7.2×10^{-3}	yes
OLLA (tuned target)	+2.14	[+1.44, +2.84]	1.9×10^{-5}	yes
Multi-OLLA	+0.20	[−0.48, +0.87]	1.0×10^{0}	no
Fixed	−0.88	[−1.44, −0.33]	7.2×10^{-3}	no
FixedPU (class oracle)	0.00	[−0.00, 0.00]	1.0×10^{0}	no
Context table	−0.88	[−1.44, −0.33]	7.2×10^{-3}	no
Bandit (global)	+1.32	[+0.67, +1.97]	7.2×10^{-3}	yes
Bandit (delay class)	+1.70	[+1.07, +2.33]	1.0×10^{-4}	yes

TABLE XI
M4 (GOODPUT, RTT 0): RESIDUAL A2C VS. EACH COMPARATOR.

Comparator	Margin	90 % CI	Holm p	Beats
Baseline	+7.07	[+6.30, +7.83]	1.5×10^{-38}	yes
OLLA	+0.15	[−0.41, +0.72]	1.0×10^{0}	no
OLLA (tuned target)	−0.90	[−1.50, −0.30]	8.2×10^{-2}	no
Multi-OLLA	−0.77	[−1.35, −0.20]	1.3×10^{-1}	no
Fixed	−1.73	[−2.23, −1.24]	1.7×10^{-8}	no
FixedPU (class oracle)	−1.04	[−1.25, −0.83]	4.9×10^{-15}	no
Context table	−1.73	[−2.23, −1.24]	1.7×10^{-8}	no
Bandit (global)	+0.15	[−0.41, +0.72]	1.0×10^{0}	no
Bandit (delay class)	+0.51	[−0.03, +1.04]	5.9×10^{-1}	no

APPENDIX C
ORIGINAL CALIBRATION AND RE-CALIBRATION DETAILS

Limits of the original calibration. A code review before the final comparison found two limits, which the re-calibration study removes.

1) *The original Context table was found by a two-pass coordinate ascent started from the all-zero table, on 10 calibration traces.* The search space has $7^3 = 343$ tables (seven offsets in three bins), and the search did not examine all of them.
2) *The original FixedPU table was separable.* For each of the nine global offsets one run was made, and each of the six classes took its own best offset from the per-class metrics. The assembled six-entry table was never evaluated as a whole table.
3) *The re-calibration study removes both limits.* Fixed (9 candidates), the Context table (343 tables) and the tuned-target OLLA (30 settings) are searched exhaustively on 500 fresh calibration traces (seeds 600001–600500). FixedPU is a separable grid followed by joint refinement (at most three sweeps). A combined table over slack × delay class × speed class × busy HARQ processes (0 or ≥ 1; $3 \cdot 2 \cdot 3 \cdot 2 = 36$ cells) is searched by coordinate ascent from 4 starts. The first start is the *composite* table

$$\mathrm{comp}(b, d, v, u) = \min\big(2, \max(-4, P(d, v) + X(b) - F)\big),$$

with F the re-calibrated Fixed offset, $P(d, v)$ the re-calibrated FixedPU table and $X(b)$ the re-calibrated Context table, all in integer CQI steps. Cells with fewer than 200 calibration decisions are not searched and keep their composite value. The tables are hashed before any evaluation trace is run, and calibration and evaluation values are reported side by side.

Calibration set. 500 traces (seeds 600001–600500) with their own CRN draws, disjoint from the M1 evaluation traces. Every table was written to a file and hashed before the evaluation traces were run.

Single-cue rules. The re-calibrated Fixed rule selects one global offset of −2 CQI steps (the same value as the original Fixed rule). The re-calibrated Context table is the best of all tables over three slack bins (−4, −4, −2). The re-tuned OLLA uses target 0.01 with a 1.0 dB step. The re-calibrated FixedPU is a separable search followed by joint refinement.

Combined table. Table XII gives the calibrated offsets of the combined table. Cells with fewer than 200 calibration decisions (marked *) were not searched and keep the composite value of Section V. The search ran from 4 starts (the composite table and three random tables) with at most 6 sweeps each. 3 starts converged; one reached the cap while still improving (score 0.6087), and the selected start S2 scored 0.6088. The search took 24.6 h over 21 sweeps.

TABLE XII
CALIBRATED COMBINED TABLE: CQI OFFSET PER SLACK BIN b (1: TIGHTEST), DELAY CLASS d, SPEED CLASS v (1: SLOWEST) AND BUSY-HARQ INDICATOR u. *: SPARSE CELL, COMPOSITE VALUE.

		$v=1$		$v=2$		$v=3$	
b	d	$u=0$	$u=1$	$u=0$	$u=1$	$u=0$	$u=1$
1	1	−3*	−3*	−4*	−3	−4*	−4*
1	2	−3*	−3*	−4*	−4	−4*	−4
2	1	−3*	−4	−4*	−4	−4*	−2
2	2	−3*	−4	−4*	−3	−4*	−3
3	1	−1	−1	−2	−1	−2	−1
3	2	−2	−1	−2	−1	−3	−2

Calibration against evaluation. Table XIII compares each re-calibrated rule's score on the calibration traces with its on-time ratio on the evaluation traces.

TABLE XIII
RE-CALIBRATED RULES: CALIBRATION OBJECTIVE AND EVALUATION ON-TIME RATIO.

Rule	Calibration	Evaluation
Combined table	0.6088	0.6086
Context table (re-cal.)	0.5982	0.6040
Fixed (re-cal.)	0.5953	0.6014
FixedPU (re-cal.)	0.5914	0.6022
Composite (start of the search)	0.6003	–

APPENDIX D
PRE-REGISTERED ROBUSTNESS PROGRAMME

Table XIV lists the pre-registered checks of Sections VI-B–VI-F with the short codes used in the released pre-registration files, the claims ledger and the code. Figure 11 shows how the verdict of the confirmatory replication (NR2) is determined.

APPENDIX E
REGISTRY SUMMARY

Table XV summarises the case registry by campaign version. A case is one configuration evaluated on its own traces;

TABLE XIV
PRE-REGISTERED ROBUSTNESS PROGRAMME. CODES REFER TO THE RELEASED PRE-REGISTRATION FILES. EACH TEXT WAS FROZEN BEFORE ITS STAGE RAN; NR2 WAS ADDED AFTER THE OTHER RESULTS WERE KNOWN. "FRESH": TRACES OR SEEDS NOT USED IN ANY EARLIER ROUND.

Check (code)	Question	Data	Decision rule	Status
Action analysis (G1)	What does the policy do, and where?	logged M1/M3 decisions, all seeds	descriptive; a mechanism sentence needs a bin $\geq 5\,\%$ of decisions and a pattern in every seed	run
Warm-up check (W1)	Is the gap to OLLA a cold-start effect?	M1 traces, OLLA warmed up for 200 TTIs	Claims A and B as in M1; the abstract quotes the smaller gain	run
Re-calibration study (G6)	Does the result survive stronger comparators?	500 fresh calibration traces; M1 evaluation traces	Holm over 10 comparators; Claims A and B, and beating the combined table; seed-level rule against the strongest comparator	run
Non-oracle base ablation (NR1)	Is the class oracle in the base rule needed?	fresh seeds, non-oracle Fixed base; M1 evaluation traces	Holm over nine; WIN = beats all	run
Operating-point check (G10)	Does it hold at other operating points?	two neighbouring cells, fresh traces; single-cue rules re-calibrated per cell	Holm over nine per cell; ALL / MIXED / NONE	run
Speed breakdown, timing (A1, A3)	By user speed; inference cost	M1 traces; timing on one CPU	descriptive	run
PF-timing demonstration (A6)	Is the multi-process HARQ gain a PF-timing artefact?	100 fresh traces, three schemes	expectation stated in advance	run
Confirmatory replication (NR2)	Does the ablation replicate on fresh data?	fresh seeds, fresh evaluation traces, frozen re-calibrated tables	trace-level and seed-level rules against the combined table	run
Operating points, non-oracle recipe (NR-G10)	Does the recommended recipe hold at other operating points?	the two G10 cells and traces, fresh seeds; single-cue rules frozen from G10	Holm over nine per cell; ALL / MIXED / NONE	run

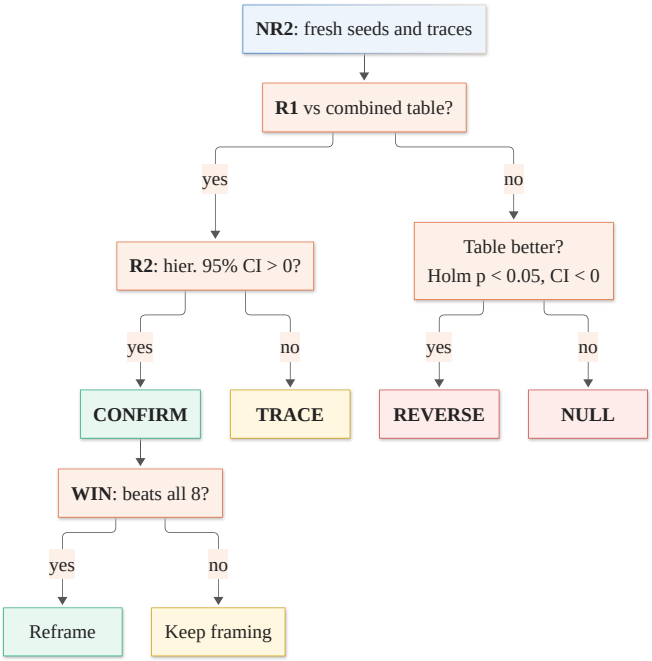


Fig. 11. Pre-registered verdicts of the confirmatory replication (NR2) against the frozen combined table. R1 and R2 are the rules of Table IV; the REVERSE branch uses the Holm-adjusted test and the upper bound of the trace-level interval. WIN (beating every comparator of the family) is reported with every verdict; a reframing of the paper is considered only after CONFIRM together with WIN.

a row is one scheme evaluated in a case. The full registry is in the supplement. The 9 cases of the robustness checks (Appendix D) are kept in a separate ledger and counted in the project-wide correction (76 cases, 126 scheme evaluations in total).

TABLE XV
CASE REGISTRY BY VERSION: CASES, SCHEME EVALUATIONS (ROWS), CASES WITH A VALID WIN, AND SUPERSEDED CASES.

Version	Cases	Rows	WIN	Superseded
v9	18	18	0	0
v10	15	15	2	0
v11	10	17	0	0
v12	2	2	0	0
v13	3	6	2	0
v14	6	12	3	0
v15	3	6	0	3
v16	6	20	1	4
v17	4	12	2	0
all	67	108	10	7

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